

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Artificial Intelligence Application Development and Jira

Practical – VII(TASK)

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1.

Area (Sq.Ft)	Price (Lakhs)
600	32
800	40
1000	49
1200	58
1400	68
1600	79

Find the regression equation.

Predict the price of a 1500 sq.ft. house.

Predict the price of a 2500 sq.ft. house.

CODE:

```
import pandas as pd
from sklearn.linear_model import LinearRegression

data = pd.DataFrame({
    'Area': [600, 800, 1000, 1200, 1400, 1600],
    'Price': [32, 40, 49, 58, 68, 79]
})

X = data[['Area']]
y = data['Price']

model = LinearRegression()
model.fit(X, y)

print("Regression Equation :")
print("Price =", round(model.coef_[0], 6), "* Area +", round(model.intercept_, 6))

house1 = pd.DataFrame([[1500]], columns=['Area'])
pred1 = model.predict(house1)

house2 = pd.DataFrame([[2500]], columns=['Area'])
pred2 = model.predict(house2)

print("Predicted Price for 1500 sq.ft. :", round(pred1[0], 2), "Lakhs")
print("Predicted Price for 2500 sq.ft. :", round(pred2[0], 2), "Lakhs")
```

OUTPUT:

Regression Equation :

Price = 0.046857 * Area + 2.790476

Predicted Price for 1500 sq.ft. : 73.08 Lakhs

Predicted Price for 2500 sq.ft. : 119.93 Lakhs

2. A company has collected the following employee data:

- Employee ID
- Employee Age
- Years of Experience
- Salary

You need to develop a Linear Regression model to predict an employee's salary.

Tasks

1. Identify the independent variable (X) and the dependent variable (Y).
2. Explain why you selected these variables.
3. Create a Pandas DataFrame using the given data.
4. Split the dataset into training and testing sets.
5. Train a Linear Regression model.
6. Predict the salary for an employee having 7 years of experience.
7. Display:
 - Slope (Coefficient)
 - Intercept
 - R2 Score
8. Plot the regression line.

CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

data = pd.DataFrame({
    'Employee_ID': [101,102,103,104,105,106,107,108,109,110],
```

```
'Employee_Age': [22,24,26,28,30,32,34,36,38,40],
'Years_of_Experience': [1,2,3,4,5,6,7,8,9,10],
'Salary': [25000,30000,35000,40000,45000,50000,55000,60000,65000,70000]
})

X = data[['Years_of_Experience']]
y = data['Salary']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=1
)

model = LinearRegression()
model.fit(X_train, y_train)

new_data = pd.DataFrame([[7]], columns=['Years_of_Experience'])
pred = model.predict(new_data)

print("Predicted Salary for 7 Years of Experience :", round(pred[0],2))
print("Slope (Coefficient) :", model.coef_[0])
print("Intercept :", model.intercept_)
print("R2 Score :", round(r2_score(y_test, model.predict(X_test)),2))

plt.scatter(X, y, color='blue')
plt.plot(X, model.predict(X), color='red')
plt.xlabel("Years of Experience")
plt.ylabel("Salary")
plt.title("Linear Regression")
plt.show()
```

OUTPUT:

```
... Predicted Salary for 7 Years of Experience : 55000.0
Slope (Coefficient) : 5000.0
Intercept : 20000.0
R2 Score : 1.0
```

